U.S. General Services Administration

Agentic AI Book Club - Session x

Reviewing Chapter 2.1 & 2.2

AI Community of Practice - 2026

Agenda

- Housekeeping
- Chapter review
- Quiz review
- Intro to next week’s chapters

Housekeeping

- **Agentic AI is an early and rapidly evolving knowledge domain**
- Knowledge Check quizzes are open-anything
- Self-study week (28th): chapter 2.3 and 2.4
- Reviewing week (5th) - Part 2 Videos for 2.2, 2.3, and 2.4. Review part 1 slides.
- CPE for attendance - in process

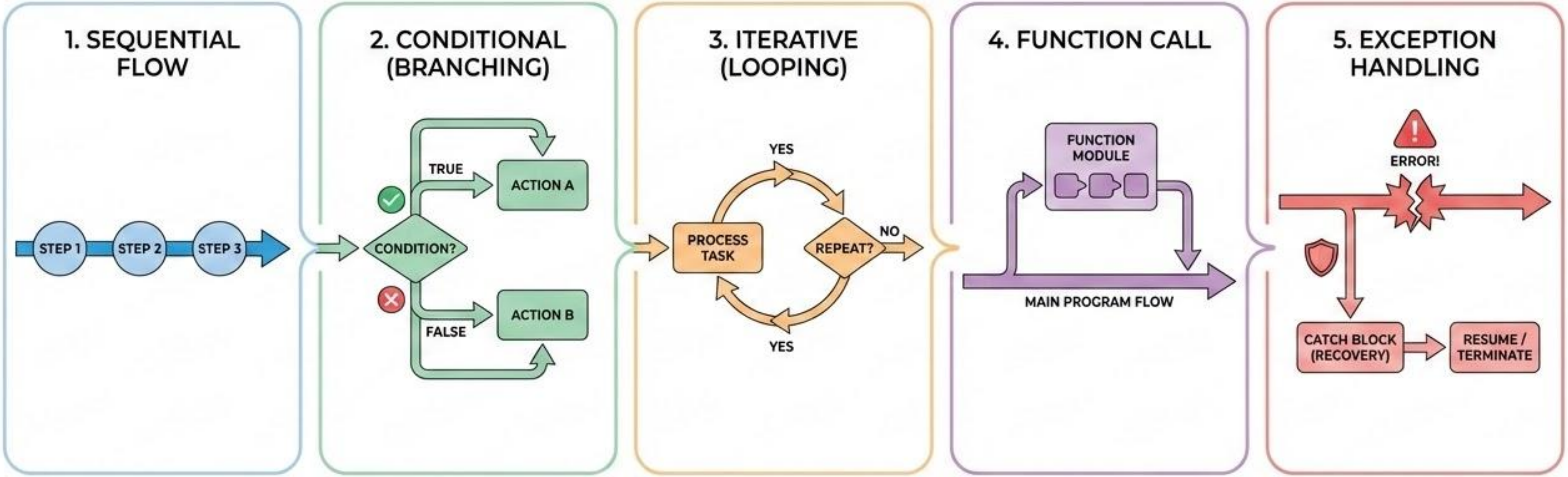
Note on CPE Credit Earning

To be awarded **1.0 CPE credit**, participants must:

- Be present for a minimum of **50 instructional minutes** (verified via platform attendance report), **and**
- Complete all **3 interactivity checks** (timed micro-quizzes delivered at unannounced intervals during the live session via the meeting chat).

Interactivity checks do **not** need to be answered correctly — submission is sufficient.

Control Flow Evolution



Linear execution, one step after another.

Decision-based paths; "If-Then-Else" logic.

Repeating tasks until a condition is met.

Jumping to modular code blocks and returning.

Managing errors, diverting flow to safe state.

Let’s review ReAct

What are the three phases in the ReAct pattern's fundamental loop that distinguish it from single-step generation?

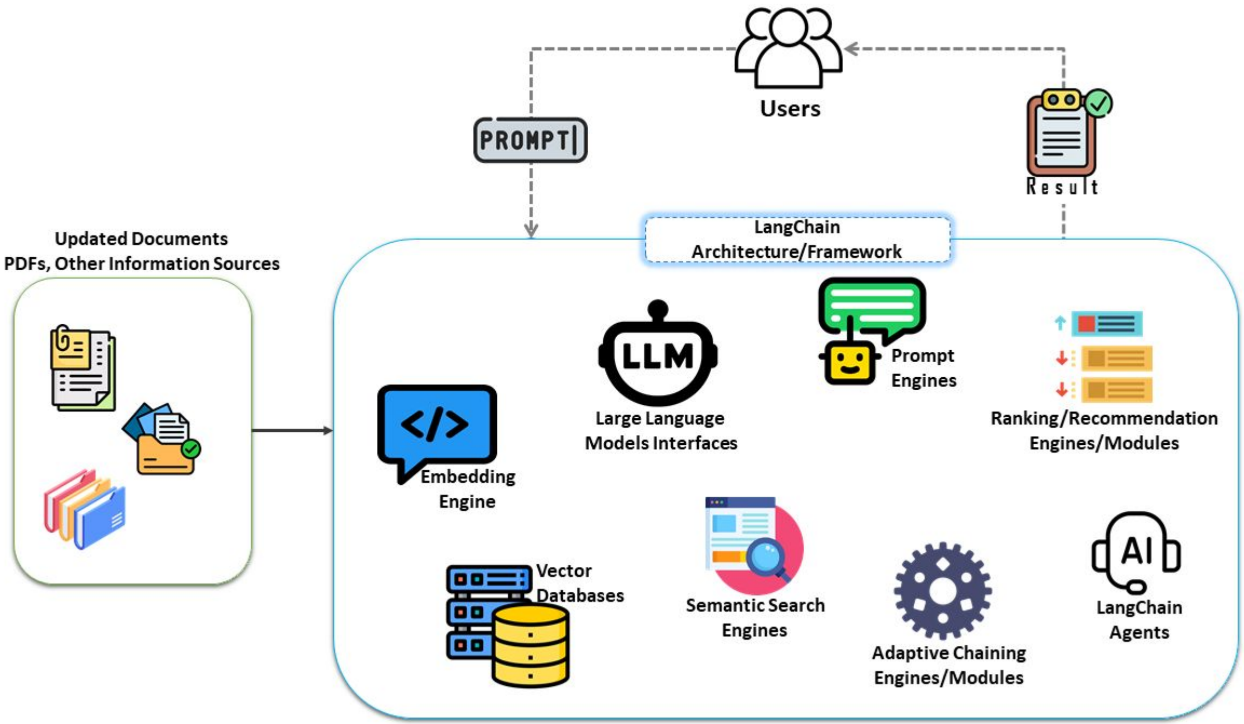
a) Plan, Execute, and Review
b) Thought, Reasoning, and Action
c) Thought, Action, and Observation
d) Plan, Act, and Review

Let’s review ReAct

What are the three phases in the ReAct pattern's fundamental loop that distinguish it from single-step generation?

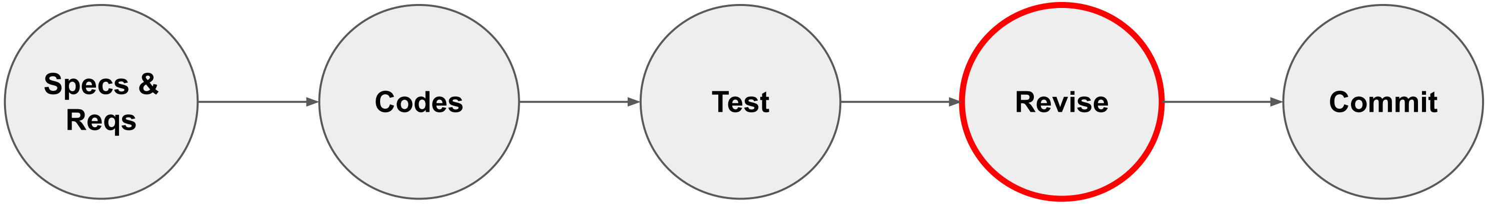
a) Plan, Execute, and Review
b) Thought, Reasoning, and Action
c) Thought, Action, and Observation
d) Plan, Act, and Review

LangChain -- Simplicity for Sequential Workflows

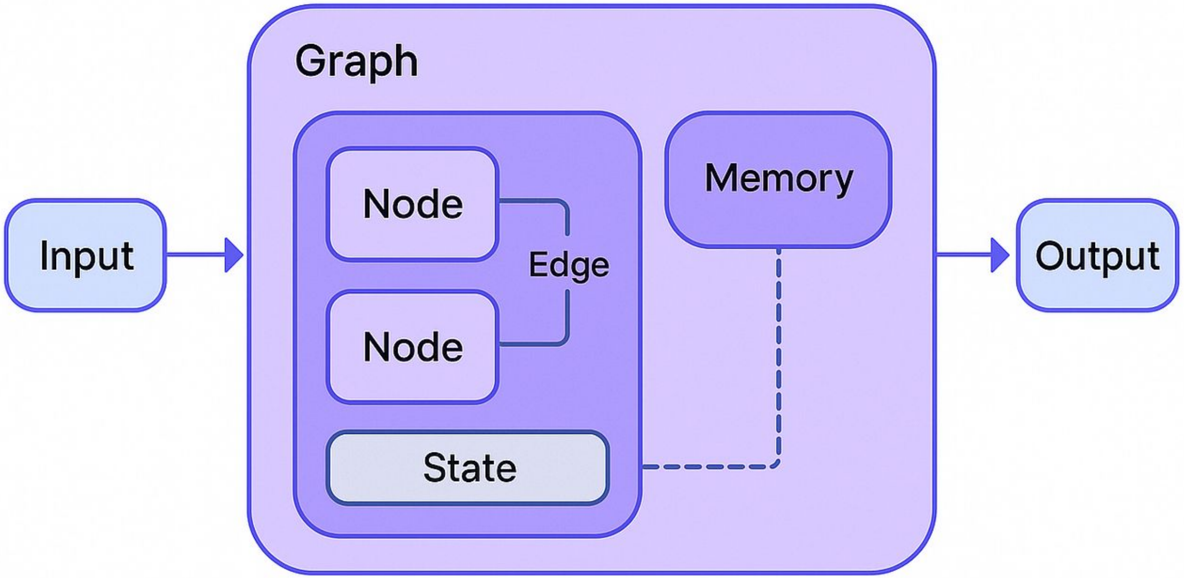


Credit: https://ai.plainenglish.io/langchain-tutorial-series-understanding-langgraph-part-5-2f123b191bc2

Case study: App Dev with Langchain

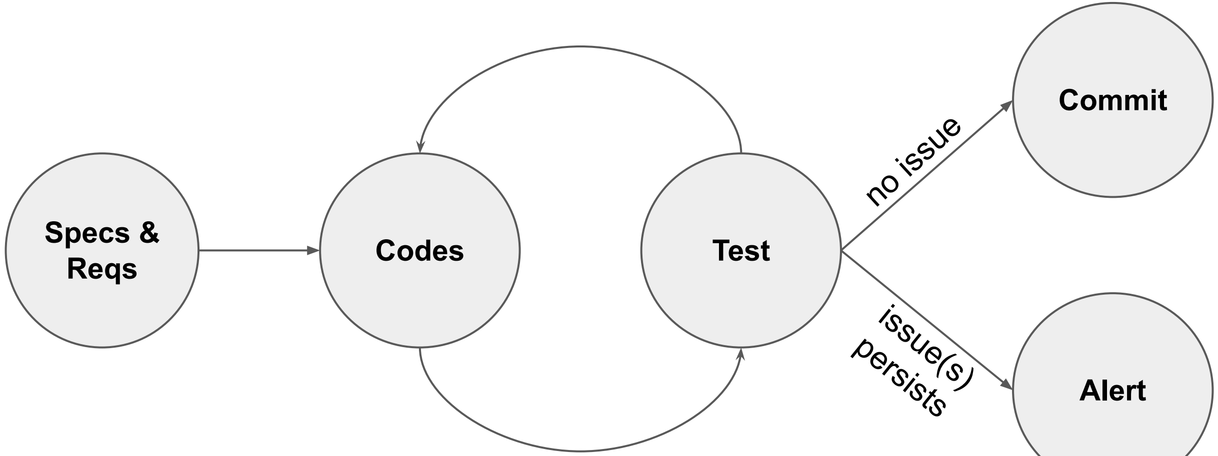


LangGraph -- Graphs for Complex State and Cycles

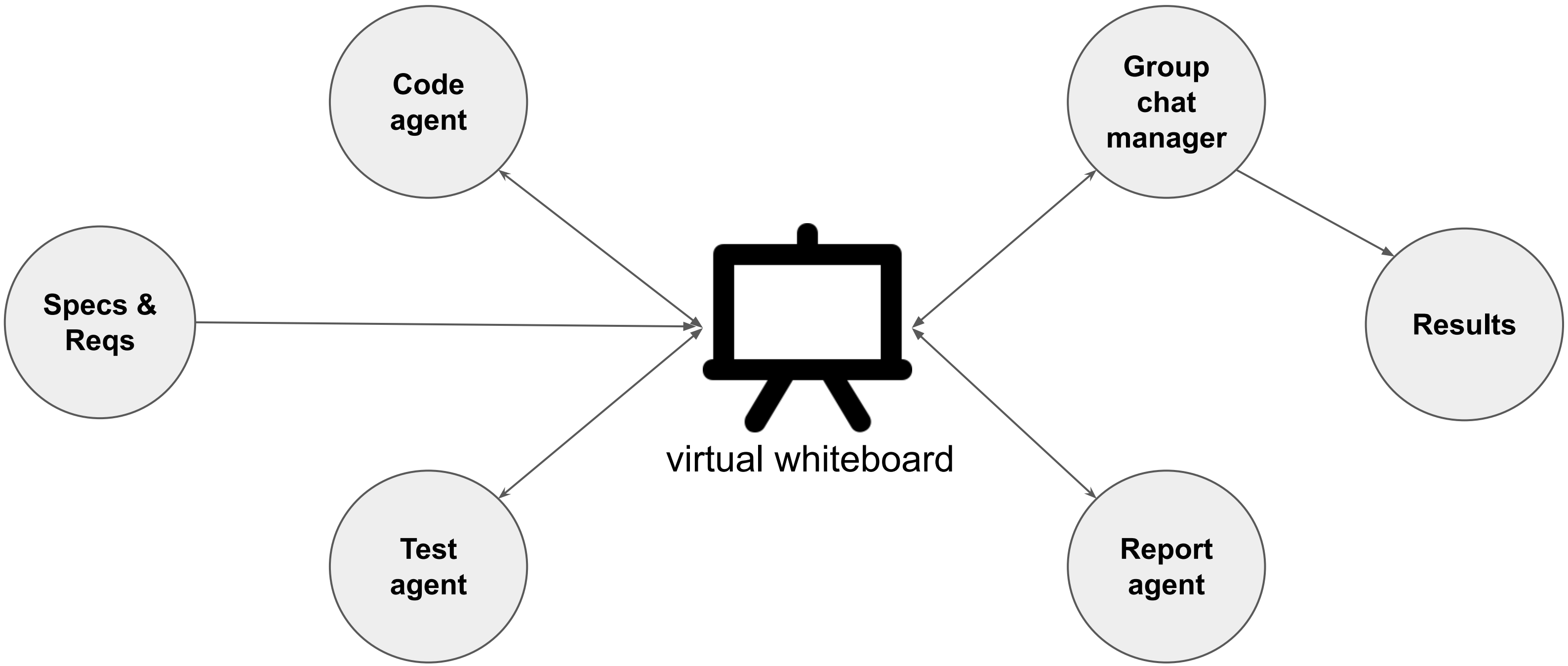


Credit: https://ai.plainenglish.io/langchain-tutorial-series-understanding-langgraph-part-5-2f123b191bc2

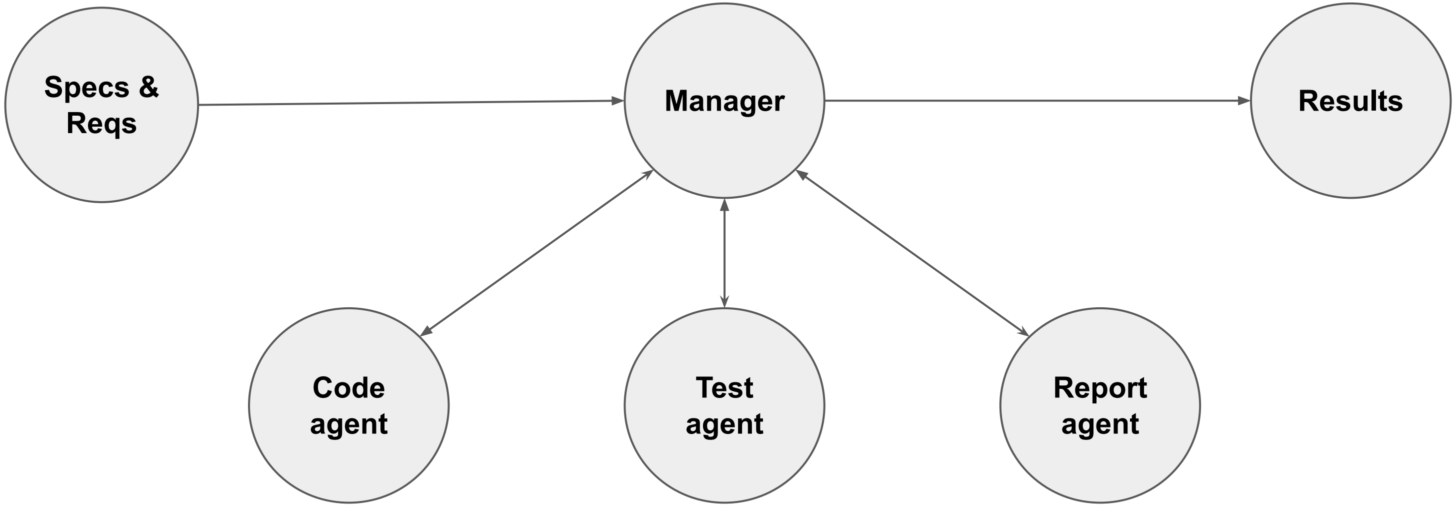
Case study: App Dev with LangGraph



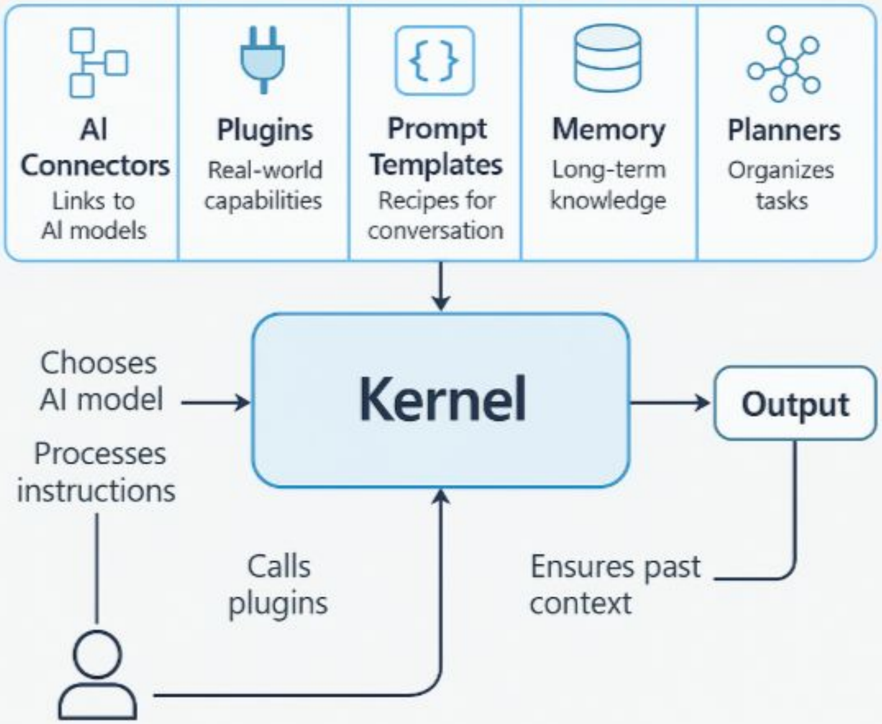
Case study: App Dev with AutoGen



Case study: App Dev with AutoGen

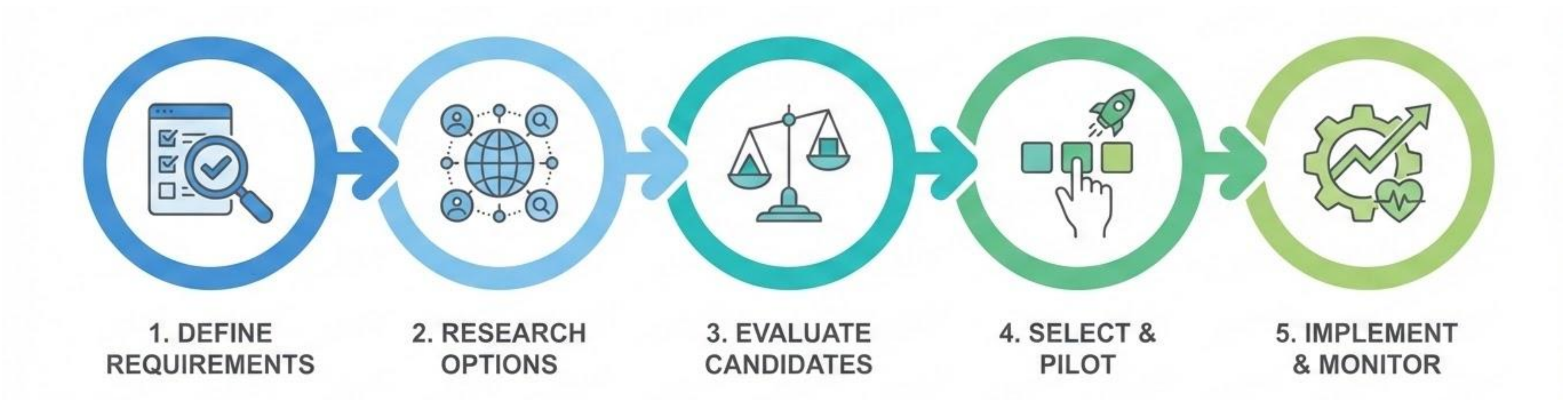


Semantic Kernel -- Orchestration Hub

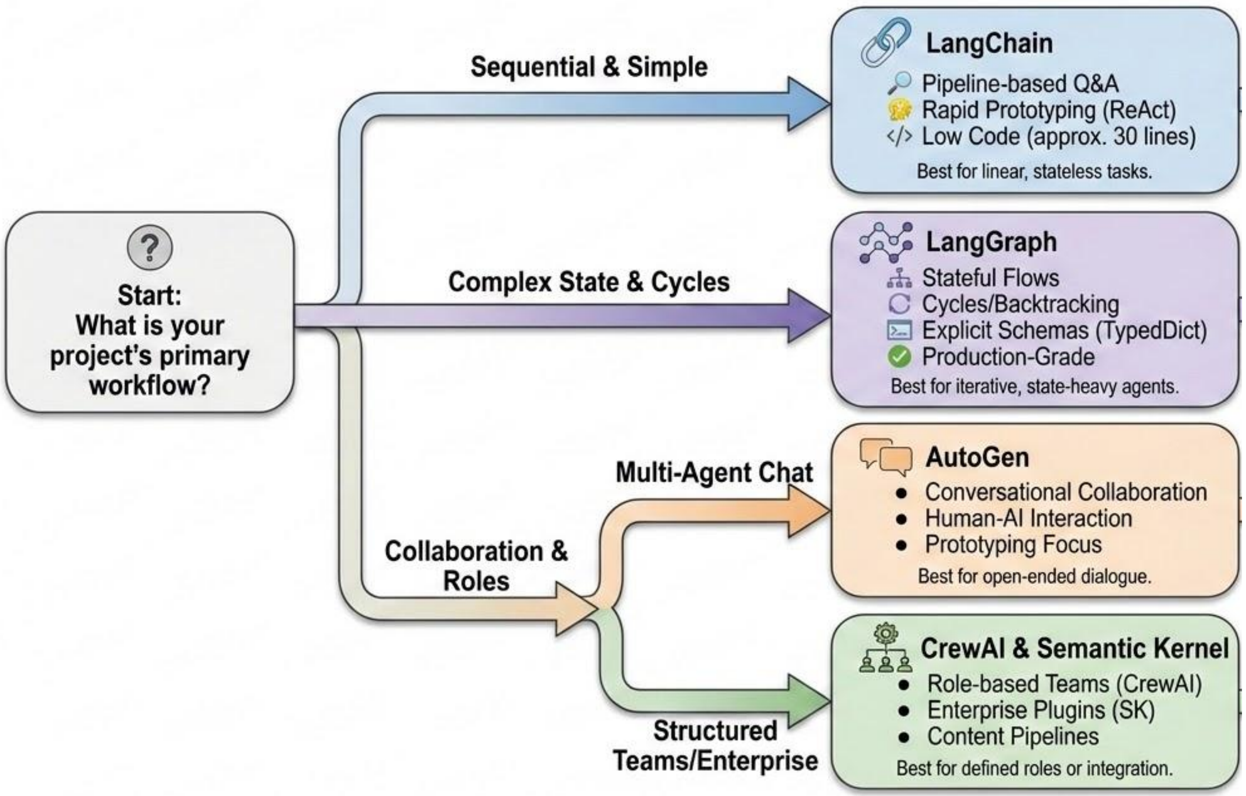


Read more: https://medium.com/microsoftazure/step-by-step-guide-to-develop-ai-multi-agent-system-using-microsoft-semantic-kernel-and-gpt-4o-f5991af40ea6

Framework Selection



Framework Quick-Reference Decision Map



Knowledge Check

Your team is building a simple FAQ chatbot that retrieves answers from a knowledge base. The sequential workflow is: receive question, search database, retrieve articles, generate answer, and return response. Which framework is probably best?

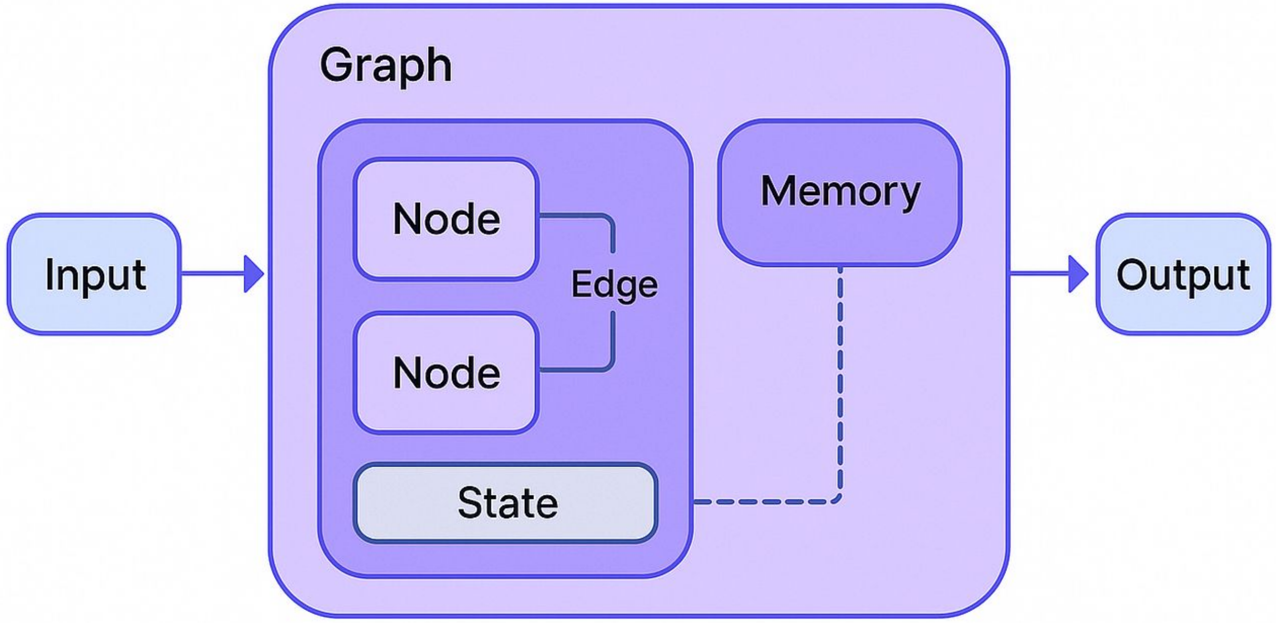
A) LangChain
B) LangGraph
C) AutoGen
D) CrewAI

Knowledge Check

Your team is building a simple FAQ chatbot that retrieves answers from a knowledge base. The sequential workflow is: receive question, search database, retrieve articles, generate answer, and return response. Which framework is probably best?

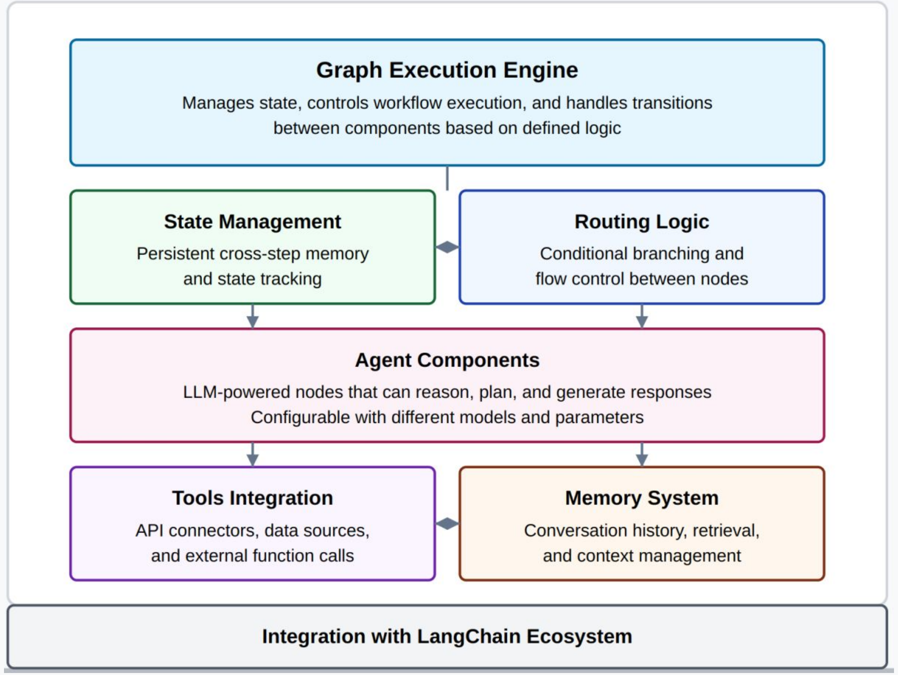
A) LangChain
B) LangGraph
C) AutoGen
D) CrewAI

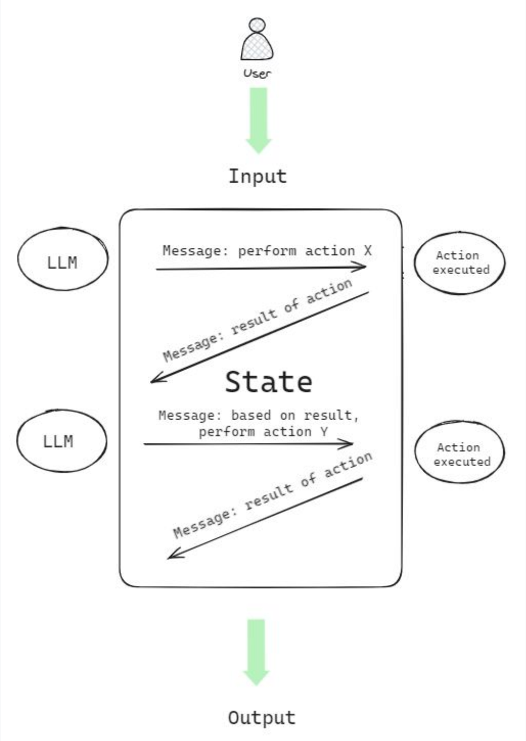
The LangGraph Building Blocks



State -- More Than Just Chat History

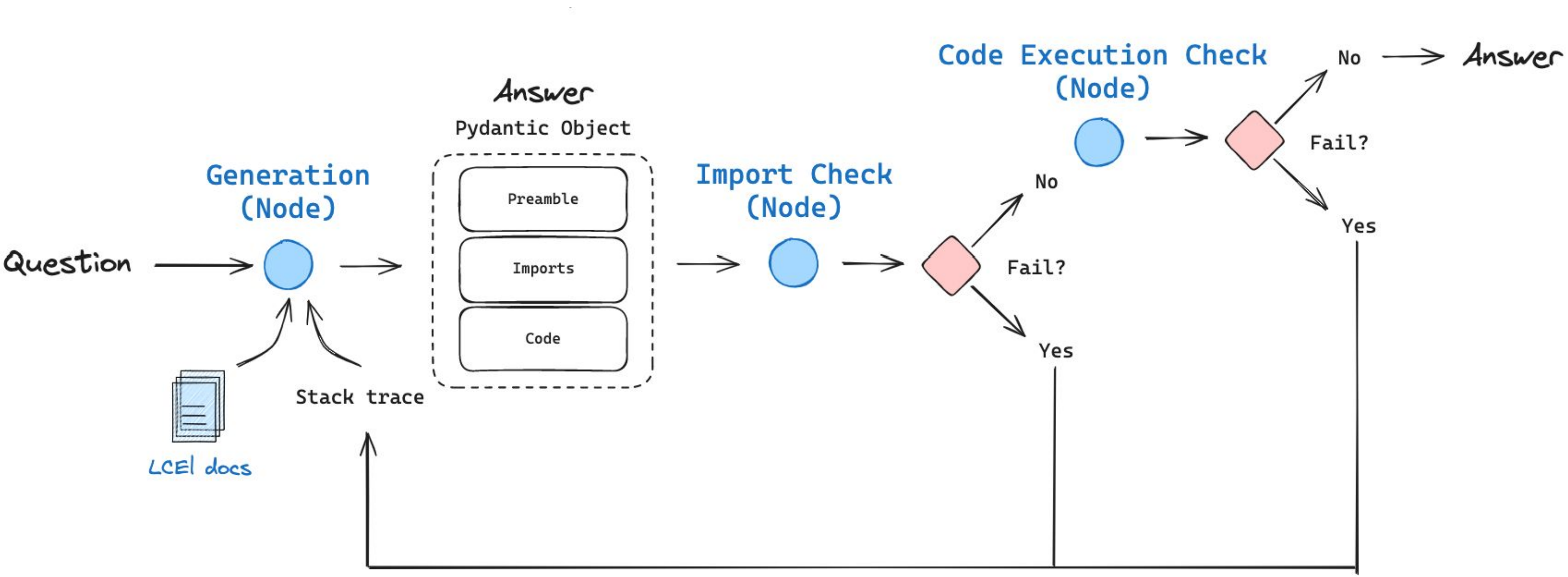
LangGraph Architecture





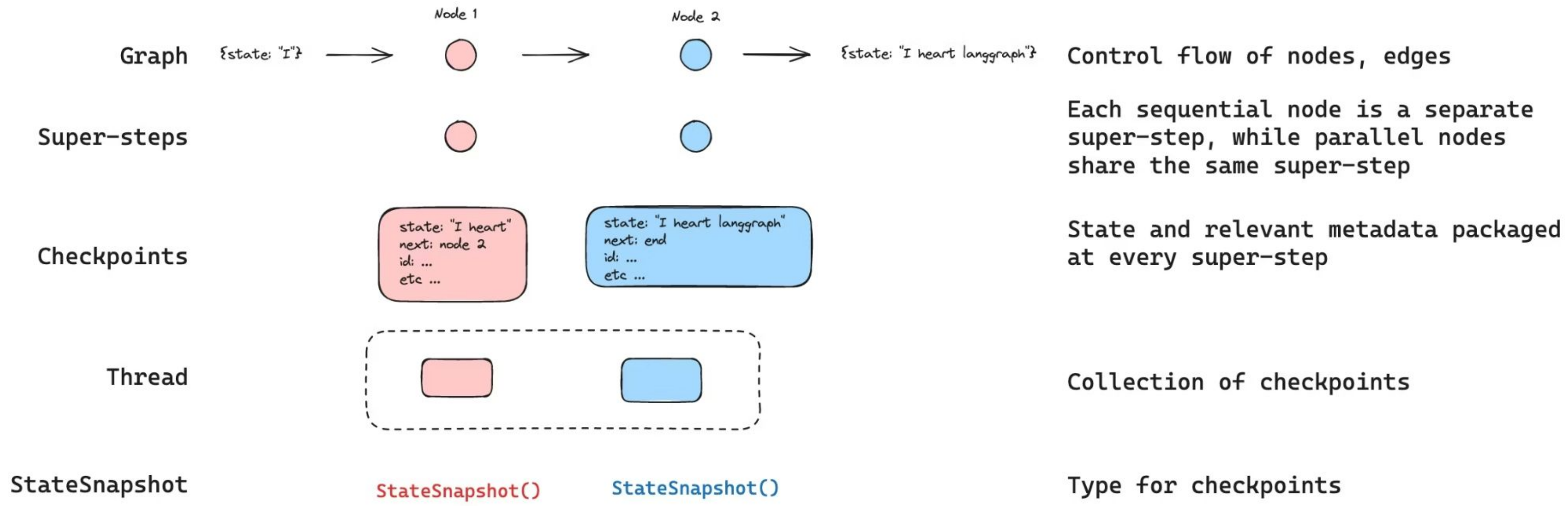
Credit: https://blog.scottlogic.com/2024/05/13/langgraph-cycling-through-llm-applications.htm
https://www.linkedin.com/pulse/understanding-langchain-langgraph-practical-guide-daghan-lemi-acay-p2iwc l

Cycles -- The Killer Feature



Credit: storage.ghost.io

Checkpointing -- Pause, Resume, Time-Travel

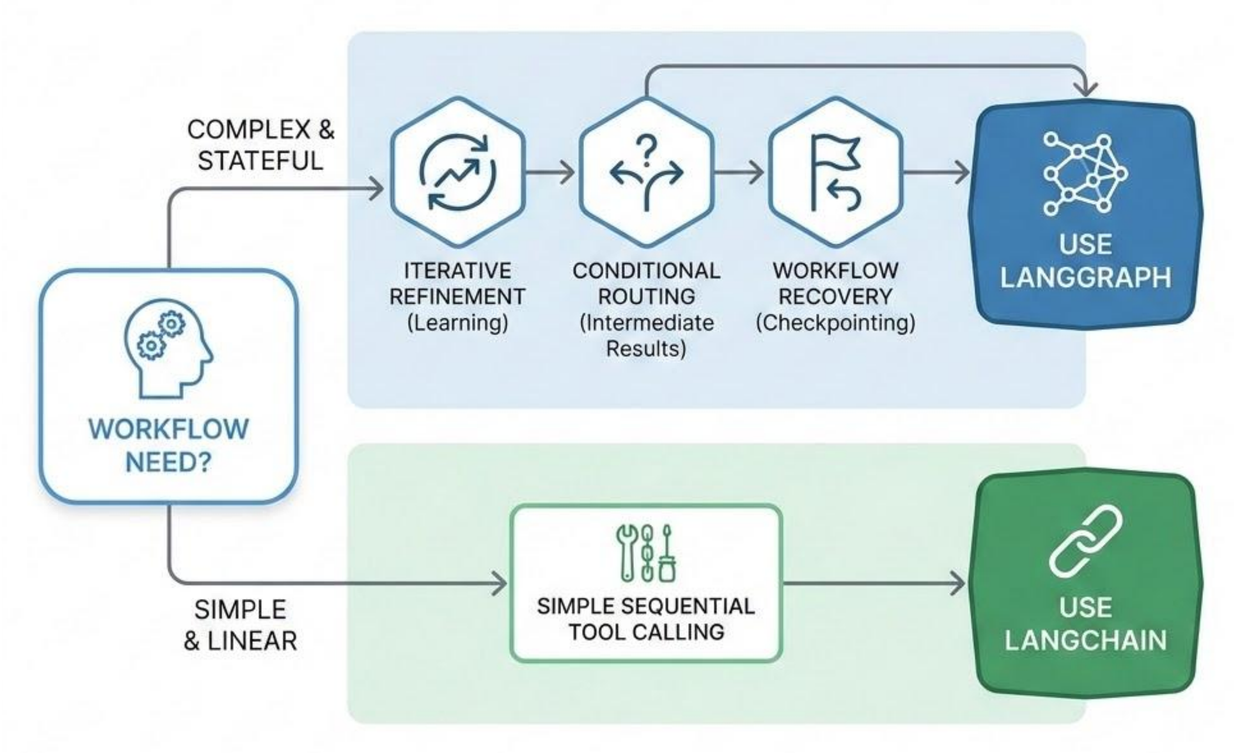


Credit: https://docs.langchain.com/oss/python/langgraph/persistence

When NOT to Use LangGraph

- Simple Q&A with search needs no graph architecture
- Sequential workflows: ~15 lines in LangChain vs. ~60 in LangGraph
- State schemas and node wrappers add development overhead
- Premature optimization trades productivity for hypothetical flexibility
- Start simple; migrate to graphs when evidence demands it

LangGraph vs Langchain Decision



Knowledge Check

You are designing a customer service agent where 70% of queries answer FAQs (simple lookup→respond), 20% escalate complex issues (classification→escalate), and 10% require iterative user-feedback refinement (answer→feedback→refine→repeat). Your manager recommends LangGraph as "best practice." What is the most accurate evaluation?

a) LangGraph is questionable for this workflow
b) LangGraph fits because it represents current best practices
c) Any workflow with multiple sequential steps requires LangGraph's graph capabilities
d) Use LangGraph now to prevent costly future rewrites

Knowledge Check

You are designing a customer service agent where 70% of queries answer FAQs (simple lookup→respond), 20% escalate complex issues (classification→escalate), and 10% require iterative user-feedback refinement (answer→feedback→refine→repeat). Your manager recommends LangGraph as "best practice." What is the most accurate evaluation?

a) **LangGraph is questionable for this workflow**
b) LangGraph fits because it represents current best practices
c) Any workflow with multiple sequential steps requires LangGraph's graph capabilities
d) Use LangGraph now to prevent costly future rewrites

Next Chapters

- **Chapter 2.3: LangChain**
  The chapter covers agent types, tool integration patterns, and recognition of when workflows exceed LangChain's linear model and require migration to LangGraph.

- **Chapter 2.4: MultiAgent Frameworks**
  This chapter explores two fundamentally different approaches to multi-agent coordination: AutoGen's message-driven conversational architecture and CrewAI's organizational structure model. It examines the trade-offs between conversational flexibility and reproducibility, along with patterns for composing multi-agent systems with specialized single-agent frameworks.